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Science, Technology, and Society

Essay 4 – Option B

05/12/2026

The Manichean Machine: Warfare as the Crucible of Modern Technoscience

The relationship between warfare and the development of science and technology extends far beyond military funding or rapid innovation during wartime. Modern war has fundamentally shaped the production of knowledge itself. Since World War II, scholars have argued that the “computerization of society” emerged largely as a byproduct of the “computerization of war” (Edwards 65). Military priorities during the Cold War influenced not only scientific research and university institutions, but also broader ideas about human cognition, communication, and control. Warfare did not simply accelerate science; it reorganized the institutions, concepts, and technologies that define modern technoscience.

One of the clearest examples of this relationship is the rise of the “iron triangle,” the close alliance between universities, private industry, and the military. Before the 1940s, American universities generally maintained a distinction between “pure” scientific inquiry and military application. World War II dramatically blurred these boundaries. Through the Office of Scientific Research and Development (OSRD), the federal government invested enormous sums into wartime research,

encouraging collaboration across disciplines and centralizing scientific work around military goals (Edwards 47).

Stanford University became a key example of this transformation under Frederick E. Terman, who aligned the university's research with military priorities to expand its funding and influence (Leslie 58). He created elite "steeples of excellence" funded largely through military contracts in electronics and microwave research. Researchers adapted their work to military needs, while the military incorporated emerging technologies of centralized command and control. By the 1950s, defense-related funding in engineering far exceeded traditional university support. This helped establish the Cold War scientific regime, where the federal government quietly shaped technological development through military spending (Mirowski and Sent 650).

Beyond transforming institutions, warfare also changed the way knowledge itself was understood. Peter Galison describes the emergence of operations research, game theory, and cybernetics as "Manichean sciences" developed through the military need to predict and control an enemy during World War II. In this framework, the enemy was increasingly treated not simply as a human opponent, but as part of a larger mechanical system whose behavior could be statistically calculated and anticipated. (Galison 230–252).

Norbert Wiener's work on the anti-aircraft predictor clearly shows this shift, as he treated both pilots and gunners as parts of a single system of communication, prediction, and feedback (Galison 240). This blurred the distinction between humans and machines, contributing to what Edwards calls "cyborg discourse." During the Cold War, the human mind itself came to be understood as an

information-processing system shaped by hierarchical control and feedback loops. These ideas were strongly influenced by military communication and command structures. In this sense, the cognitive revolution reflected not only scientific progress but also the logic of wartime technological systems (Edwards 199–222).

The technologies produced through this military-scientific alliance also embodied specific forms of political power. Paul Edwards uses the term “closed-world discourse” to describe the worldview that emerged through Cold War defense projects such as SAGE, an early computerized command-and-control system integrating radar, computation, and human decision-making into a unified defense network. SAGE imagined the United States as something that could be enclosed, monitored, and managed through continuous technological control (Edwards 262).

These systems were based on the belief that global conflict could be understood as a rule-governed problem manageable through computation and technical superiority. As a result, they privileged technical and standardized forms of knowledge over local or experiential understanding (Edwards 41). Even when these systems failed in practice, such as Operation Igloo White in Vietnam, where an “electronic battlefield” of sensors failed to stop guerrilla forces, the broader belief in centralized technological control remained intact (Edwards 139).

In this sense, technologies are not politically neutral tools. As Langdon Winner argues, certain technologies inherently support particular social and political arrangements. Accepting systems such as the nuclear-industrial complex also means accepting broader structures of technocratic

authority, including what Winner describes as a “techno-scientific-industrial military elite” (Winner 130).

A major debate within STS concerns whether military funding distorted science or redirected it. Paul Forman argues that Cold War military funding pushed physics toward applied techniques and practical “gadgeteering,” encouraging technical effectiveness at the expense of deeper theoretical understanding. Daniel Kevles, however, argues that physics was not distorted so much as reorganized into new areas made intellectually attractive through national security priorities. Scientists were not passive victims of military power, but active participants in reshaping their discipline. Nevertheless, even Kevles acknowledges that by the late 1950s, military agencies had become the central source of funding for physics research, making military patronage the defining condition of the field (Oreskes 18–23).

The relationship between war and technoscience continues into the twenty-first century through “dual-use” technologies that operate in both civilian and military contexts (Rappert et al. 721). Contemporary systems such as AI-driven surveillance, predictive algorithms, and autonomous weapons all have roots in military research and counterinsurgency strategies. Even the foundations of the internet emerged from ARPA-funded projects intended to secure military communication systems during the Cold War.

Today, conflict zones increasingly function as testing grounds for new forms of surveillance and automated warfare. At the same time, universities, corporations, and governments continue to collaborate through systems justified in the name of “national security” (Mirowski and Sent 652-

655). As these systems expand, distinctions between human and machine, simulation and reality, become increasingly unstable. Video games, military simulations, and virtual reality training environments normalize technologically mediated warfare while distancing audiences from its consequences (Rappert et al. 727–729).

Ultimately, warfare does not simply accelerate scientific development; it reshapes the institutions, concepts, and technologies through which modern society operates. Across the frameworks discussed in this essay, from Galison’s “Manichean sciences” to Edwards’ “closed-world discourse,” warfare and technoscience emerge as deeply interconnected systems that continually reinforce one another. Modern technologies such as computing, the internet, and artificial intelligence cannot be fully understood outside the military contexts that shaped their development. Their histories are tied not only to innovation and efficiency, but also to systems of surveillance, control, and geopolitical power that continue to define the modern world.

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